THE LAWS OF SOFTWARE TEAMS

Conway's law

Organizations design systems that copy their communication structure

swipe →

Ivan Rykov

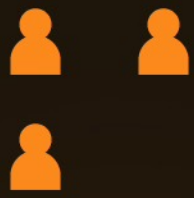
1968 · 8 people, 2 compilers

Teams in Same shape out



COBOL · 5 people

- phase 1
- phase 2
- phase 3
- phase 4
- phase 5



ALGOL · 3 people

- phase 1
- phase 2
- phase 3

Melvin Conway, “How Do Committees Invent?”, Datamation, 1968

02 THE LAW

Organizations design systems that copy their own communication structure

Conway’s law · paraphrased from the 1968 paper

architecture



Teams that don't talk ship code that doesn't fit



Glue code piles up in between

Microsoft · Windows Vista

The org chart predicted the bugs

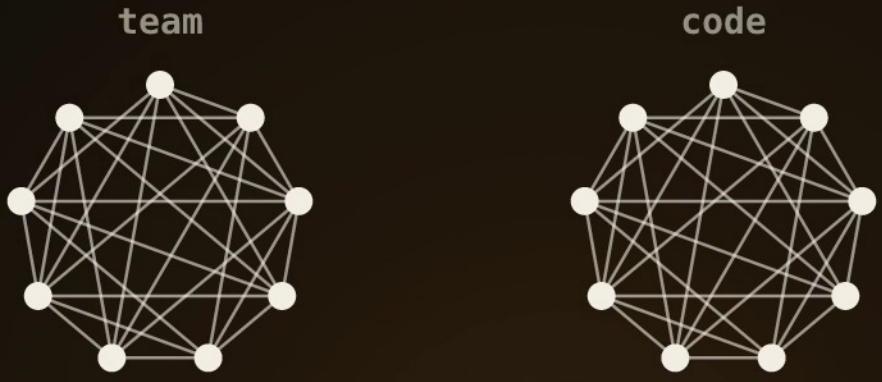


3,404 binaries · 50M+ lines of code

86% precision from org structure, better than churn, complexity, coverage

Nagappan, Murphy & Basili, ICSE 2008 · 1 dot ≈ 10 binaries

Your code is a mirror



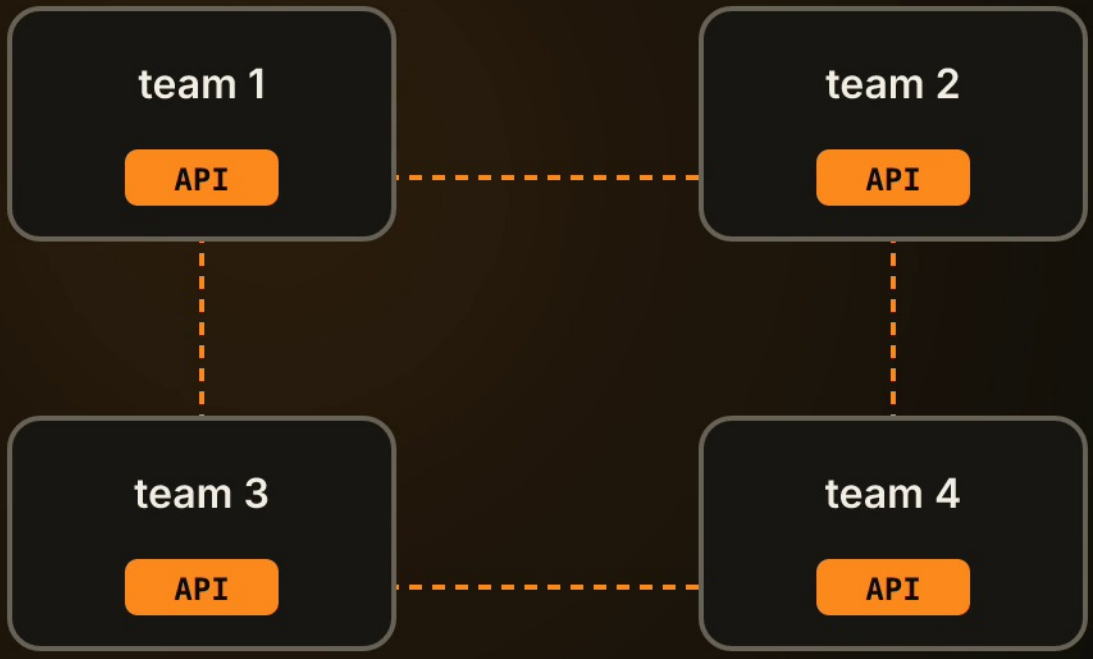
tightly connected → tangled



loosely connected → modular

MacCormack, Rusnak & Baldwin · HBS & MIT · diagram illustrative

Flip it: design the teams first



Amazon, ~2002: teams talk only through service interfaces

the "inverse Conway maneuver" · via Steve Yegge's 2011 platforms memo

Flo: logic moved to where the team is



each team: 1 iOS · 1 Android · 2-4 backend

app store release — every few weeks

backend-driven releases

20-30 a day

via manager.dev

Want a different architecture?



Draw a different org chart

TAKEAWAY

Want a different architecture? Draw a different org chart

Sources: Melvin Conway, Datamation, 1968 · Nagappan, Murphy and Basili, 2008 · manager.dev on Flo

next: Dunbar's number